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School of Computing

Bachelor of Computer Science

PRJ-303

Spring 2026

Blockchain-Based Insurance

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Application for the Project

Written with LLM assistance.

1. Project Identification

1.1 Reference Number:

1.2 Problem statement

Traditional insurance markets rely on centralized, opaque databases to manage policy lifecycle and ownership. While effective for basic administration, these systems suffer from a lack of portability, high administrative overhead for verification, and a "black box" environment where policyholders have no independent, immutable proof of their coverage.

Blockchain technology, specifically Non-Fungible Tokens (NFTs), offers a solution by providing a transparent, immutable, and verifiable ledger of ownership. However, mass adoption is currently blocked by the **"Complexity Barrier."** The general public is largely unwilling or unable to manage cryptographic private keys, navigate the volatility of gas fees, or interact with decentralized wallets (e.g., MetaMask). Furthermore, storing sensitive personal insurance data directly on a public blockchain presents significant privacy and regulatory risks.

1.3 Project Title: Blockchain-Based Insurance

1.4 Key Words:

Insurance, Blockchain, Smart Contracts, Ethereum, Immutable Ledger, Decentralized Application (DApp), Non-Fungible Tokens(NFTs).

2. Aims, Goals, Objectives and scope of the Project

2.1 Aim of the Project:

To develop and deploy a hybrid **web 2.5** insurance application that brings the power of blockchain technology to the industry with minimal to no complications related to adopting blockchain into the real world.

2.2 Goals of the Project:

The project aims to:

- Standardize insurance policies as tradable, verifiable digital assets (NFTs).
- Abstract away blockchain complexity through familiar "Web2" interfaces (email/password and fiat payments).
- Securely handle policyholder privacy by separating sensitive document storage from public ownership records.

2.3 Objectives of the Project:

To achieve the project goal, the following objectives will be pursued:

- **Non-Custodial Managed Wallet System:** Development of a client-side encryption module that allows users to generate Ethereum-compatible keys, encrypt them with a passphrase, and store them securely on a centralized database.
- **Fiat-to-NFT Pipeline:** Integration of the **Stripe API** to process credit card payments, which triggers a backend webhook to mint or transfer an insurance NFT to the user's managed wallet.
- **Policy Tokenization:** A Solidity Smart Contract (ERC-721) deployed on a testnet (e.g., Sepolia) that records policy ownership, expiry dates, and cryptographic hashes of policy documents.
- **User & Admin Dashboards:** A frontend interface for users to view their "NFT Policies" and a basic admin panel for the insurance company to issue or revoke NFTs based on off-chain events.
- **Data Privacy Architecture:** Implementation of a metadata strategy where sensitive policy details are kept off-chain, using only hashes and tokenized URLs on the blockchain.

2.4 Scope of the Project:

a. System scope

- **Wallet Generation Engine:** Automatically creates a unique public/private key pair on the client side using libraries like viem.
- **Security & Encryption Module:** Implements a standard AES-256 layer to turn the private key into a JSON keystore before it leaves the user's browser.
- **Payment & Minting Bridge:** A middleware that listens for **Stripe Webhooks**. Once a successful fiat payment is confirmed, the system triggers the "Mint" or "Transfer" function on the blockchain.
- **NFT Lifecycle Management:**The smart contract handles the state of the policy identifying if it is *Active*, *Expired*, or *Revoked* and ensures only the verified owner can hold the token.
- **Metadata Management:** Dynamically generates JSON responses for the tokenURI, allowing the NFT to display different images or statuses (e.g., a "Red Badge" if a policy lapses) based on database values.
- **Hash Verification Service:** Provides a utility to compare the hash stored in the NFT against a physical PDF document to prove the document hasn't been tampered with.

b. User scope

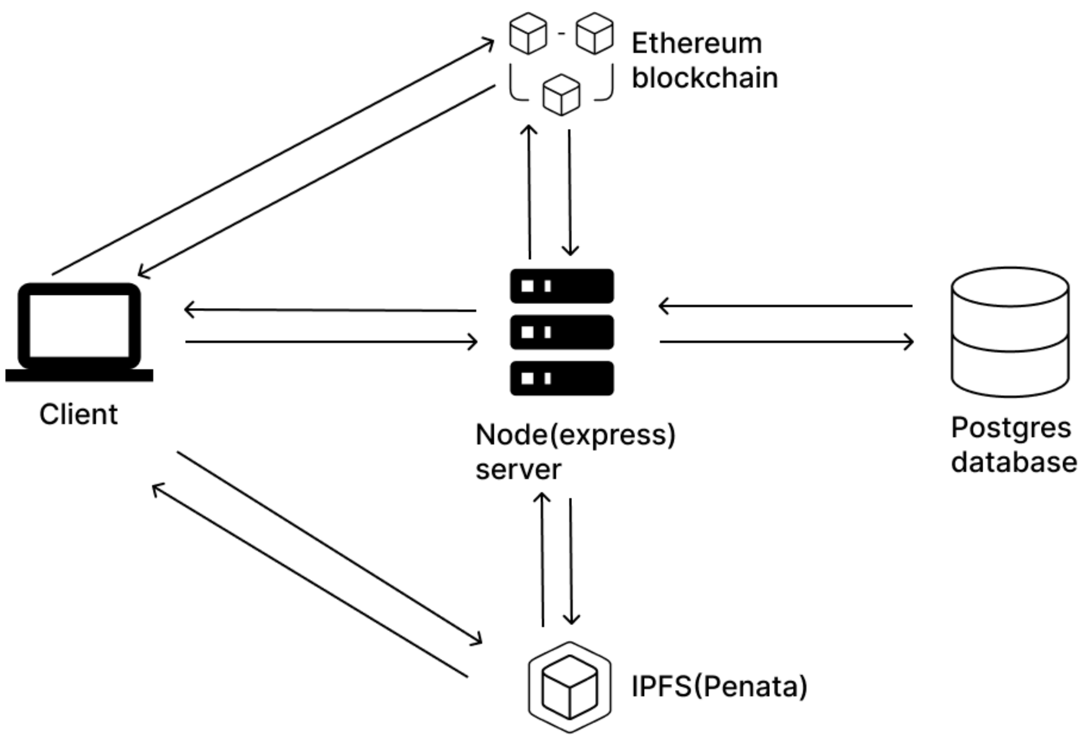
The system is designed to serve multiple stakeholders within the Bhutanese insurance ecosystem:

1. The Insurance Administrator
    - **Policy Templating:** Create new types of insurance (Life, Travel, Car) and set the basic metadata.
    - **Manual Intervention:** Review submitted claims and manually mark them as "Approved" or "Denied" in the database.
    - **Revocation:** Trigger a contract function to "Burn" or deactivate an NFT if a customer disputes a payment or a policy is found to be fraudulent.
    - **Custom Minting:** For unique requirements, the admin can manually draft a specific policy, hash it, and mint a custom NFT directly to a user's address.

2. The Policyholder (Customer)
    - **Onboarding:** Create an account and set a "Wallet Password" (which encrypts their key).
    - **Purchase:** Browse available insurance products and pay with fiat.
    - **Dashboard Access:** View all purchased policies in a "Digital Vault" (essentially an NFT gallery).

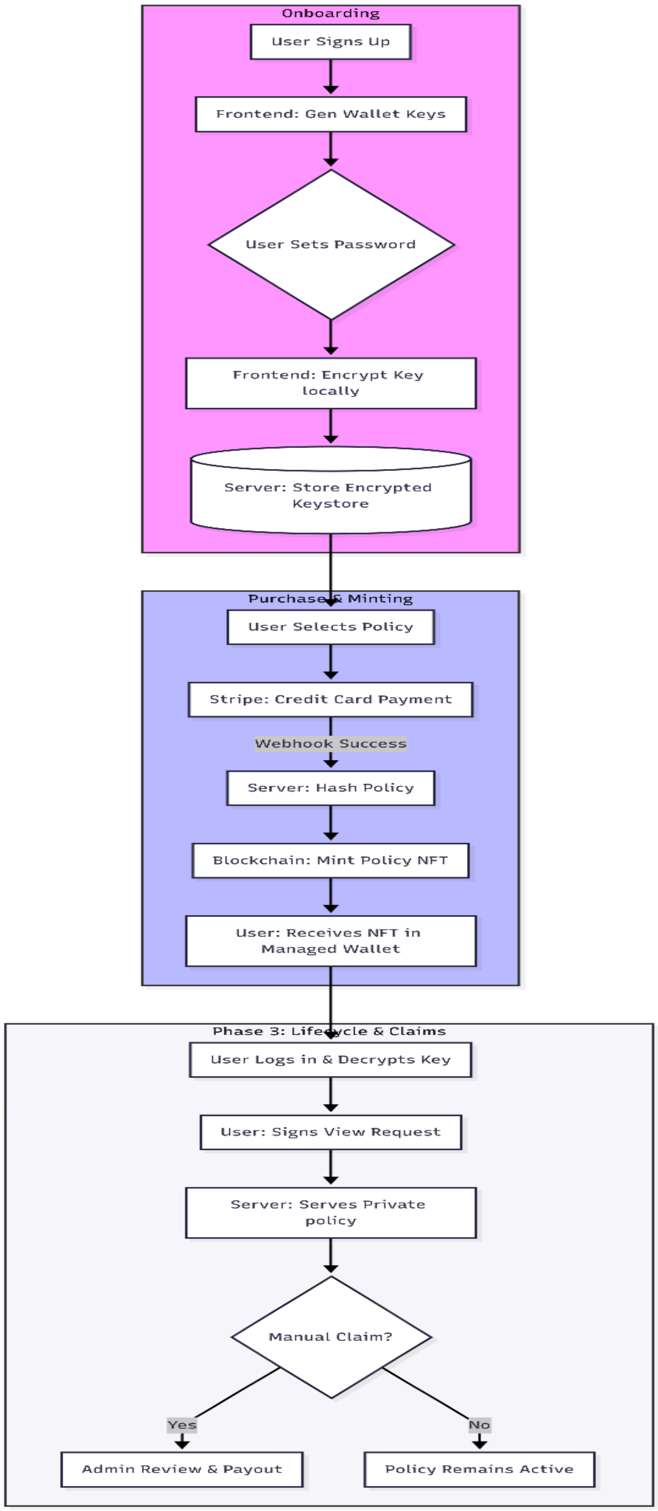
- **Verification:** Export the policy NFT's metadata or hash to prove coverage to third parties (e.g., proving travel insurance at an embassy).
- **Claim Initiation:** Click a "File a Claim" button that sends an off-chain notification and the NFT ID to the admin.

3. System Architecture

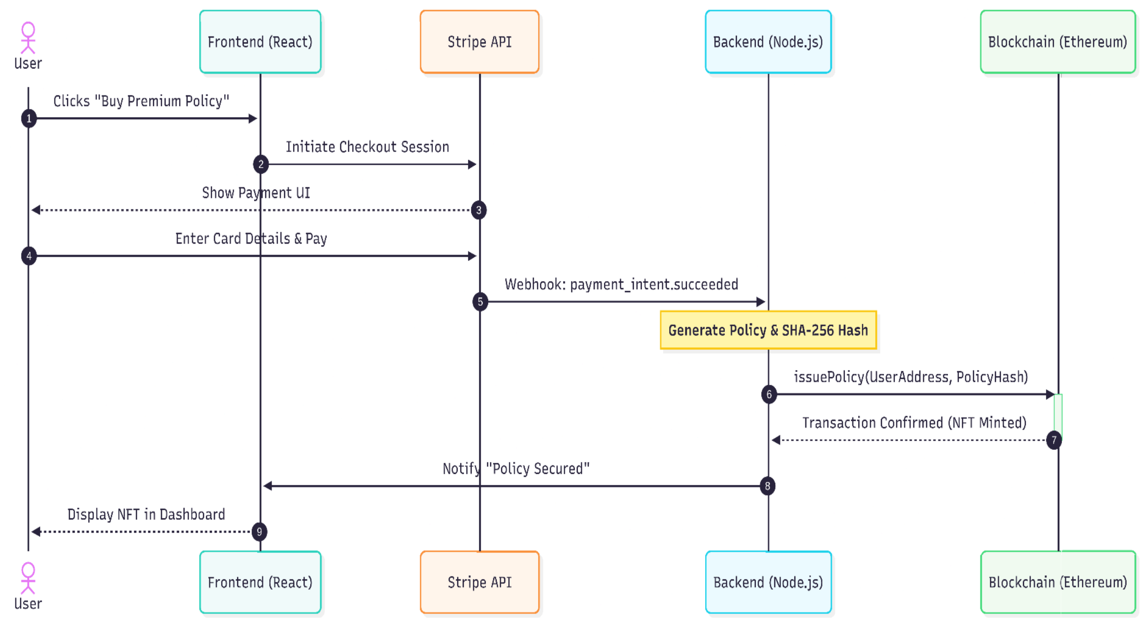


4. Workflow

High Level workflow:



Technical workflow:



4. Project Background

Insurance systems traditionally operate using centralized databases and manual administrative workflows. While this approach has been functional for decades, it presents several limitations such as lack of transparency, slow verification processes, high administrative overhead, and increased susceptibility to fraud. Policyholders must fully trust insurance providers, as they have no independent, tamper-proof method to verify policy ownership or authenticity.

In recent years, blockchain technology has emerged as a powerful tool for improving trust, transparency, and data integrity. Blockchain's immutable ledger ensures that once information is recorded, it cannot be altered without detection. These properties make blockchain particularly suitable for applications involving ownership verification and long-term record preservation, such as insurance policies.

However, direct adoption of blockchain in insurance faces significant barriers. Typical blockchain systems require users to manage cryptographic private keys, interact with decentralized wallets, pay transaction fees, and understand complex concepts such as gas fees and token standards. For the general public, these requirements create a high entry barrier, limiting real-world adoption. Additionally, storing sensitive insurance data directly on public blockchains raises serious privacy and regulatory concerns.

To address these challenges, this project proposes a **hybrid Web 2.5 blockchain-based insurance system**. The system combines the usability of traditional Web2 applications—such as email-based authentication and fiat payments—with the security and immutability of blockchain technology. Insurance policies are represented as **Non-Fungible Tokens (NFTs)**, enabling verifiable ownership and lifecycle management, while sensitive documents are securely stored off-chain.

By abstracting blockchain complexity from end users and adopting a privacy-preserving architecture, the proposed system demonstrates how blockchain can be practically integrated into the insurance domain with minimal disruption. The project focuses on policy issuance, ownership verification, and fraud resistance rather than full automation of claim settlements, making it suitable as a realistic and scalable prototype for the Bhutanese insurance ecosystem.

4.1. Literature Review:

1. Overview of Traditional Insurance Systems

Traditional insurance systems operate on centralized architectures where policy issuance, claims processing, and data storage are managed internally by insurance companies. Even globally digitized insurers such as Allianz and AXA rely on centralized databases for policy administration (Allianz Group, 2023; AXA Group, 2023).

In centralized insurance systems:

- Policy ownership is recorded in internal databases
- Claims are manually reviewed and approved
- Policy documents are issued as PDFs or printed certificates
- Verification requires contacting the issuing company

While digital transformation has improved efficiency, the trust model remains institutional rather than cryptographic (Eling & Lehmann, 2018).

2. Insurance Systems in Bhutan

Bhutan's insurance market is primarily served by:

- Royal Insurance Corporation of Bhutan Limited (RICBL)

- Bhutan Insurance Limited (BIL)

Both institutions provide services such as life, motor, travel, and health insurance (RICBL, 2023; BIL, 2023). Their systems are digitized to an extent, allowing online premium payments and downloadable policy documents. However, ownership and verification remain centralized.

Policies are:

- Stored in proprietary databases
- Issued as digital documents (PDF)
- Verified internally by company representatives
- Modified or revoked by administrative authority

There is no publicly verifiable cryptographic proof of ownership layer in these systems. This structure aligns with conventional centralized insurance architecture identified in global literature (Eling & Lehmann, 2018).

3. Blockchain Technology as a Trust Mechanism

Blockchain technology provides a decentralized ledger where data is recorded immutably. Once information is written to the blockchain, it cannot be modified without consensus from the network. This characteristic directly addresses integrity and trust issues present in centralized systems.

In insurance applications, blockchain enables independent verification of ownership and policy existence without relying solely on the insurer's internal database. Shared ledgers improve auditability and reduce disputes, making blockchain a suitable foundation for policy verification systems.

4. Tokenization of Insurance Policies Using NFTs

Non-Fungible Tokens (NFTs), based on standards such as ERC-721, allow representation of unique digital assets on blockchain networks (Buterin, 2014). Unlike fungible tokens, NFTs cannot be duplicated, making them ideal for representing individual insurance policies. Each

NFT can encode ownership, validity period, and policy state, ensuring uniqueness and traceability.

Tokenizing insurance policies as NFTs allows insurers to standardize policy representation while enabling policyholders to independently verify ownership. Research suggests that NFT-based asset representation improves transparency and reduces record manipulation risks in ownership-driven domains.

Unlike centralized PDF certificates used by RICBL and BIL, NFT-based policies provide cryptographic proof of ownership that is publicly verifiable without relying solely on institutional validation.

5. Smart Contracts for Policy Lifecycle Management

Smart contracts are self-executing programs deployed on blockchain networks that enforce predefined rules. In insurance systems, smart contracts can manage policy issuance, expiry, and revocation based on encoded conditions.

Rather than fully automating claim payouts which require complex real-world verification, smart contracts are more effectively used for policy lifecycle management. This hybrid approach balances automation with necessary human oversight, aligning with practical constraints in real-world insurance environments.

6. Privacy Challenges in Blockchain-Based Systems

Although blockchain ensures transparency, it is not suitable for storing sensitive personal or legal documents directly. Public blockchains expose stored data to all participants, raising privacy and regulatory concerns, especially in insurance contexts.

To address this, many studies recommend hybrid architectures where sensitive data is stored off-chain while cryptographic hashes are recorded on-chain. Hash-based verification preserves document integrity without exposing confidential information. This approach forms the basis of the proposed system's data privacy architecture.

7. Bridging Blockchain and Traditional Payment Systems

One major obstacle to blockchain adoption is the requirement for cryptocurrency usage. Most users are unfamiliar with digital wallets and reluctant to manage private keys or volatile crypto assets. Integrating fiat payment systems provides a practical solution.

By linking traditional payment gateways with blockchain minting processes, insurance systems can maintain familiar user experiences while still benefiting from blockchain-backed ownership verification. This integration significantly lowers adoption barriers and aligns blockchain solutions with real-world user expectations.

8. Research Gap Addressed by the Proposed System

Existing blockchain-based insurance solutions often assume full user engagement with Web3 tools and cryptocurrency ecosystems. From reviewing traditional insurance systems, Bhutanese insurers (RICBL & BIL), and decentralized insurance research, a gap is identified:

- Traditional systems lack cryptographic transparency.
- Fully decentralized insurance platforms suffer from usability and adoption barriers.
- Emerging markets require solutions that balance familiarity with innovation.

There is limited research on hybrid Web 2.5 insurance architectures that:

- Maintain email/password authentication
- Accept fiat payments
- Abstract blockchain complexity
- Provide NFT-based ownership
- Preserve data privacy through off-chain document storage and on-chain hash verification

The proposed project addresses this gap by introducing a **Web 2.5 insurance model**, where blockchain operates behind the scenes while users interact through conventional web interfaces.

The system demonstrates a balanced approach that combines usability, security, and privacy, making it suitable as a realistic prototype for modern insurance systems rather than a purely experimental blockchain application.

5. Requirements

5.1. Functional Requirements.

- **User Registration:** The system must allow users to create an account via email and password.
- **Wallet generation:** Upon signup, the system must generate a unique Ethereum-compatible public/private key pair on the client side.
- **Key Encryption:** The system must encrypt the private key using a user-provided passphrase (AES-256) before storing the JSON keystore in the database.
- **Stripe Integration:** The system must integrate a checkout flow allowing users to pay insurance premiums using credit cards.
- **Automated Minting:** Upon receiving a Stripe "Payment Success" webhook, the system must automatically trigger a smart contract function to mint a policy NFT to the user's address.
- **Metadata Engine:** The system must serve dynamic JSON metadata for each NFT, including policy type, start/end dates, and the policy hash.
- **Document Hashing:** The system must generate a SHA-256 hash of the insurance policy and store it on-chain to ensure document integrity.
- **Policy Revocation:** The system must allow an administrator to mark an NFT as "Lapsed" or "Revoked" on the blockchain if payments stop or fraud is detected.
- **Digital Vault:** Users must have a dashboard to view their active policy NFTs and download the associated encrypted policy documents.

5.2. Non-Functional Requirements

1. Security (Highest Priority)
    - **Client-Side Privacy:** Private keys must **never** be transmitted to the server in plain text. Only the encrypted keystore blob is stored.

- **Authentication:** The system must implement JWT (JSON Web Tokens) or similar for secure session management.
- **Smart Contract Security:** The contract must use standard libraries (like OpenZeppelin) to prevent re-entrancy and unauthorized access.

2. Usability & Performance
    - **Latency:** The user interface should load fast. For blockchain actions, the UI must show a "Transaction Pending" state to manage expectations of network speed.
    - **Gas Efficiency:** The system should be deployed on a Layer 2 network (like **Polygon/Arbitrum**) to ensure transaction costs remain minimal.
    - **Responsiveness:** The dashboard must be fully responsive and accessible via mobile browsers.
3. Reliability & Availability
    - **Data Persistence:** User wallet addresses and policy hashes must be redundantly stored to prevent data loss.
    - **Uptime:** The API and frontend should maintain **99% uptime** during the demo/project testing phase.
    - **Webhook Reliability:** The backend must handle Stripe webhook retries to ensure no paid policy is ever "missed" by the minting engine.

6. Limitations.

- **Automated Claims Payouts:** This project will not use Oracles (like Chainlink) to automate claims. Claim verification requires complex real-world evidence (medical reports, police data) and will remain a manual process initiated by the company.
- **On-Chain Liquidity:** The system will not handle the storage of "Claim Reserve" funds in crypto. Payments and payouts remain strictly in fiat to avoid the regulatory complexity of acting as a crypto-exchange.
- **Legal & Regulatory Compliance:** While the project addresses privacy (GDPR-friendly hashing), it is a technical prototype and does not comply with actual insurance licensing laws or financial regulations.

- **Advanced Key Recovery:** If a user loses their encryption passphrase, the system will not support mathematical key recovery. Instead, it relies on the administrative "Revoke and Re-issue" function of the smart contract.

7. Technology

6.1. Frontend:

- Vite + React
- Viem(ethereum interface)

6.2. Development Environment:

- Foundry (Smart contract development and testing)
- VsCode(IDE)

6.3 Backend/smart contract:

- Nodejs(Express)
- Solidity Contract

6.4 Deployment:

- Development and Testing: Sepolia network.
- Production:Most probably an Ethereum layer-2 solution like Arbitrum..

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